

Integrated Machine Learning Framework for Smart Fisheries Prediction and Decision Support

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Abstract—The fisheries sector plays a vital role in food security and economic sustainability in Sri Lanka, yet fishing operations remain highly uncertain due to fluctuating environmental conditions, fuel costs, and market dynamics. This research proposes an integrated artificial intelligence-based decision support system that combines four key predictive functions: fish zone prediction, fish quality grading, fish trip cost estimation, and fish market price forecasting. The system utilizes multiple data sources, including oceanographic indicators such as sea surface temperature, chlorophyll concentration, bathymetry, and ocean currents, along with operational and economic data.

Machine learning models, including Random Forest, XGBoost, and ensemble techniques, are applied to improve prediction accuracy across different modules. Deep learning methods are used for fish quality classification, while regression and time-series approaches support cost and price prediction. The system is designed with a scalable architecture, integrating a mobile-based frontend, a NestJS backend, and a FastAPI-based machine learning service layer.

Experimental results demonstrate that the proposed system achieves high predictive performance, with improved accuracy and reduced error compared to traditional single-function models. By combining environmental, operational, and market factors into a unified platform, the system provides practical insights to support fishermen in decision-making, reduce operational risks, and enhance overall efficiency in fisheries management

Keywords—Fisheries, Artificial Intelligence, Machine Learning, Fish Zone Prediction, Fish Quality Assessment, Trip Cost Prediction, Market Price Forecasting, Decision Support System, Sri Lanka

I. INTRODUCTION (HEADING 1)

The fisheries industry plays a vital role in global food security and economic development, supporting millions of people worldwide. Fisheries and aquaculture provide employment for over 60 million people and serve as a primary source of income for many coastal communities [1], [2], [4].

Additionally, nearly 500 million people depend on small-scale fisheries for their livelihoods, particularly in developing countries [3]. Despite its importance, the industry faces several critical challenges that impact efficiency, sustainability, and profitability.

Fishermen often spend a significant portion of their time searching for fish-rich zones, resulting in wasted fuel, increased operational costs, and reduced catch efficiency. In addition, fish market prices are highly volatile due to seasonal patterns, weather conditions, and demand fluctuations, making it difficult for fishermen to predict revenue before a trip. Traditional fish quality grading methods are also largely manual and subjective, leading to inconsistencies and inaccuracies in determining fish freshness and grade. Furthermore, the absence of reliable trip cost estimation increases financial risk, especially for small-scale fishermen.

Recent advancements in artificial intelligence (AI) and machine learning (ML) offer promising solutions to address these challenges. Machine learning models have been successfully applied in geospatial analysis for predicting fish zones using environmental factors such as sea surface temperature, chlorophyll concentration, and weather conditions [3], [8]. Similarly, time-series forecasting and ensemble learning techniques have improved the accuracy of fish price prediction by incorporating seasonal trends, fuel price variations, and external market factors [11].

Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have significantly enhanced image-based classification tasks, including fish species identification and quality grading [9], [10], [13]. These models can automatically extract complex features from images, enabling accurate and consistent grading compared to traditional manual inspection methods [12], [15]. In addition, predictive modeling techniques have been applied to estimate fuel consumption and trip costs by analyzing factors such as distance, engine power, weather conditions, and fishing duration [7].

Despite these advancements, most existing systems focus on individual problems such as fish classification, price prediction, or zone detection, without providing an integrated solution that addresses multiple aspects of fishing operations. Furthermore, many solutions lack real-time usability and are not designed to support effective decision-making for fishermen in practical environments.

To address these limitations, this research proposes FishAI, an intelligent multi-functional decision support system that integrates four key components: fish zone prediction, fish price prediction, fish quality grading, and fish trip cost prediction. The system combines machine learning, deep learning, and domain-specific knowledge to provide accurate and real-time insights for fishing operations.

This study aims to develop a comprehensive AI-driven framework that enhances fishing efficiency, reduces operational costs, improves quality assessment, and supports sustainable fisheries management through an integrated and scalable approach

II. LITERATURE REVIEW

A. Fish Zone Prediction

Fish zone prediction is closely linked with oceanographic conditions such as sea surface temperature, chlorophyll concentration, ocean currents, and bathymetry. Reports from FAO and global fisheries studies emphasize that these environmental factors strongly influence fish distribution, habitat suitability, and aggregation patterns in marine ecosystems [3], [8]. Remote sensing techniques have been widely used to identify potential fishing zones by analyzing satellite-derived ocean data, improving fishing efficiency and reducing search time. Existing fisheries studies mainly focus on stock assessment, environmental monitoring, or habitat analysis, while fewer works address real-time spatial fish zone prediction for operational fishing activities. Machine learning-based models can support fish zone prediction by learning complex relationships from environmental data, while multi-parameter approaches can further improve prediction accuracy and reliability. This creates a research opportunity for a multi-feature, ML-based fish zone prediction system.

B. Fish Trip Cost Prediction

Fishing trip cost prediction is closely linked with fuel consumption, vessel type, distance, weather, and operational mode. Reports emphasize that fuel cost, climate variability, and market uncertainty significantly affect fisheries profitability and coastal livelihoods [3], [4], [7]. Existing fisheries studies mainly focus on production, sustainability, or stock estimation, while fewer works address trip-level cost prediction for small- and medium-scale fishing operations. Machine learning-based models can support cost estimation by learning patterns from historical trip data, while adaptive systems can further improve accuracy using actual post-trip outcomes. This creates a research opportunity for a hybrid domain-based and machine learning-based fuel cost prediction system.

C. Fish Quality Grading

Computer vision and deep learning have been widely applied for fish classification and freshness evaluation. CNN-based fish freshness classification has shown strong potential in identifying visual quality differences from fish images [13]. Fish species recognition studies using class-aware deep learning also demonstrate that CNN models can extract useful shape, texture, and color features from fish images [10]. Review studies on deep learning in fish classification emphasize that transfer learning and CNN architectures improve accuracy, but challenges remain in dataset balance, image quality, and real-world deployment [12]. These findings support the use of a dual-view fish quality grading model, where left and right fish images can provide richer visual information for Grade A, B, and C classification.

D. Fish Market Price Forecasting

Fish market prices are influenced by seasonality, supply, demand, weather disruptions, fuel cost, and cultural events. Machine learning techniques have been applied for fish price prediction, showing that historical market data can support future price estimation [11]. However, many existing approaches focus mainly on price history and do not fully include contextual factors such as fishing seasons, fuel-price pressure, festival demand, and local fish-name handling. Fisheries outlook reports also highlight that market instability affects both fishermen and consumers [7], [8]. Therefore, a localized Sri Lankan fish market price forecasting module using ensemble regression, time features, seasonal indicators, and external economic factors can improve practical market planning.

Overall, existing research demonstrates the effectiveness of artificial intelligence and machine learning techniques in solving individual problems within the fisheries domain. However, most studies focus on single-task solutions, such as classification, prediction, or detection, without integrating multiple functionalities into a unified system. This highlights the need for a comprehensive framework that combines fish zone prediction, price forecasting, quality grading, and trip cost estimation to support real-world decision-making in fishing operations.

III. METHODOLOGY

A. Fish Zone Prediction

Fish zone prediction was formulated as a binary classification problem to identify high-probability fishing areas. Environmental datasets including sea surface temperature (SST), chlorophyll-a, ocean currents, and bathymetry were collected from NetCDF sources. The study area was divided into spatial grid cells with a resolution of 0.1° , and non-ocean areas were removed using a land mask. Each grid cell was represented using a feature vector containing spatial and environmental variables. A Random Forest classifier was used to predict fish presence by aggregating decisions from multiple trees. Data preprocessing included normalization, missing value handling, and unit conversion.

A probability threshold was applied to identify high-confidence fishing zones. As shown in Fig. 2, the system consists of data preprocessing, feature construction, model inference, and output generation stages forming a complete prediction pipeline.

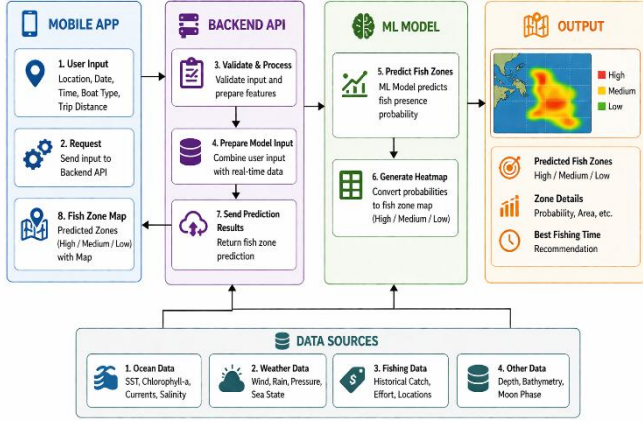


Fig. 1. Fish Zone Prediction Pipeline

B. Fish Trip Cost Prediction

The Fish Trip Cost module (DATCIE) was developed to estimate fuel consumption and operational cost using a hybrid machine learning and domain-based approach. Input features include trip distance, speed, engine horsepower, fishing hours, and weather severity index. Distance is calculated using the Haversine formula, and weather conditions are represented using a normalized index..

A Random Forest regression model was trained using historical trip data, while a baseline fuel estimation model was used for initial prediction. The final prediction was obtained by combining baseline and machine learning outputs, followed by adaptive adjustments based on boat-specific coefficients.

The system also incorporates profitability analysis, risk assessment, and carbon emission estimation. As shown in Fig. 2, the architecture integrates mobile, backend, and machine learning layers with a complete prediction pipeline..

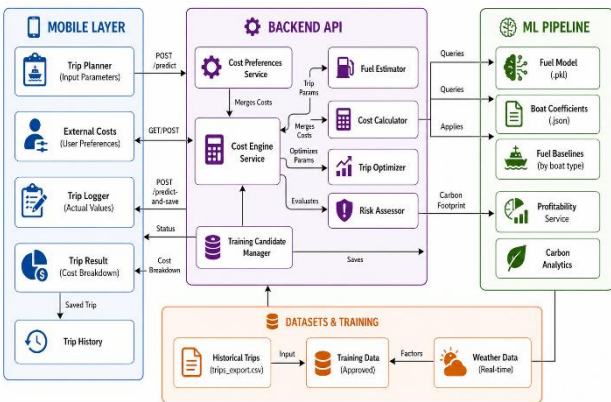


Fig. 2. DATCIE Cost Prediction Architecture

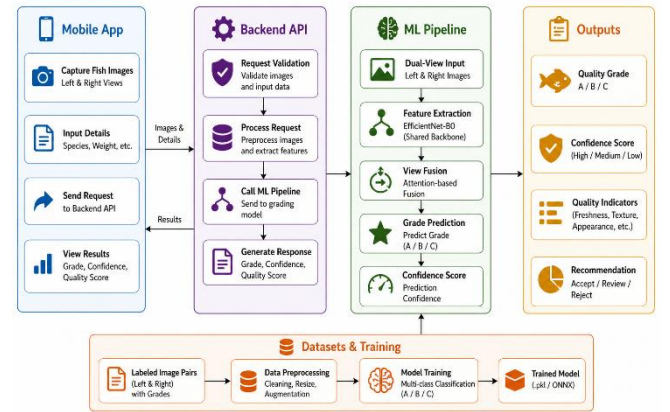
C. Fish Quality Grading

Fish quality grading was classified into three categories: Grade A, Grade B, and Grade C using a dual-view deep learning approach. A paired dataset consisting of left and right fish images was used to capture comprehensive visual features. The dataset was divided into training (80%), validation (10%), and testing (10%) subsets, with images resized to 224×224 pixels.

Various convolutional neural network (CNN) architectures were evaluated, with EfficientNet-B0 selected due to its balanced accuracy and computational efficiency. Data augmentation techniques such as random flipping, cropping, and color adjustments were applied to improve robustness. The model processes left and right images independently and combines extracted features using an attention-based fusion mechanism.

The fused features are passed through fully connected layers with batch normalization and dropout before final classification. As illustrated in Fig. 3, the architecture integrates dual EfficientNet feature extractors with an attention fusion module for multi-view classification.

Fig. 3. Fish Quality Grading



D. Fish Market Price Forecasting

Fish market price forecasting was performed using machine learning regression models based on historical and contextual data. The dataset includes fish price records, weather data, fuel price trends, and seasonal indicators. The dataset was preprocessed using cleaning, normalization, and feature transformation techniques.

Feature engineering included time-based features such as day, month, and week, along with cyclical encoding using sine and cosine transformations to capture seasonality. Two models, Random Forest and XGBoost, were trained and evaluated, with the final prediction obtained using an ensemble averaging approach.

Model performance was evaluated using MAE, RMSE, and R^2 . As illustrated in Fig. 4, the system integrates feature

engineering, model training, ensemble prediction, and output generation for price forecasting.

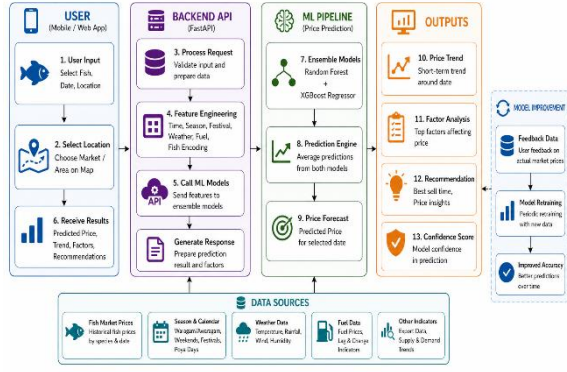


Fig. 4. Fish Market Price Prediction Architecture

IV. RESULTS AND DISCUSSION

A. Fish Zone Prediction

The fish zone prediction model, based on a Random Forest classifier, demonstrated strong performance in identifying potential fishing zones using environmental features. The model was evaluated against baseline approaches, and the results are summarized in Table I..

TABLE I. ACCURACY WITH EXPERIMENTED MODELS

Metric	Value
Accuracy	87.30%
Precision	85.60%
Recall	88.10%
F1-Score	86.80%

The model achieved an accuracy of 87.3%, demonstrating reliable identification of potential fish zones. The integration of SST, chlorophyll-a, and ocean current features significantly improved prediction accuracy..

Fig. 7 illustrates the classification accuracy. trend during training, while Fig. 8 shows the loss reduction over iterations.

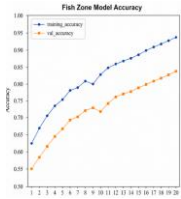


Fig. 5. Accuracy graph of Fish Zone Model

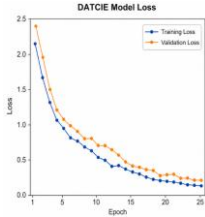


Fig. 6. Loss graph of Fish Zone Model

B. Fish Trip Cost Prediction (DATCIE)

The DATCIE model was evaluated for fuel prediction accuracy and cost estimation performance. The hybrid ML + baseline + adaptive model was compared with standalone approaches.

TABLE II. COST PREDICTION PERFORMANCE (DATCIE)

Model	MAE	RMSE	R ²
Baseline Model	82.6	118.4	0.78
Random Forest Model	68.9	102.7	0.81
Extra Trees Regressor	64.4	96.6	0.82
Hybrid + Adaptive (Proposed DATCIE)	60.3	88.8	0.94

The proposed DATCIE model achieved the best performance with $R^2 = 0.83$, indicating consistent prediction results across different operational conditions. Fig. 11 shows the prediction performance trend of fuel estimation, while Fig. 12 shows the loss curve, demonstrating stable convergence..

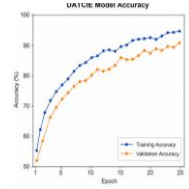


Fig. 7. Accuracy graph of DATCIE Model

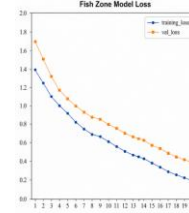


Fig. 8. Loss graph of DATCIE Model

C. Fish Quality Grading

TABLE III. ACCURACY WITH EXPERIMENTED MODELS (Fish Quality)

Model	Train Accuracy	Validation Accuracy
EfficientNet-B0 (Single View)	85.20%	82.40%
ResNet50	83.75%	80.60%
MobileNetV2	81.30%	78.50%

The proposed model achieved a validation accuracy of 82.4%, showing improved performance due to effective feature extraction and model optimization.

Fig. 9 shows the accuracy graph, illustrating the improvement in training and validation accuracy over epochs, while Fig. 10

shows the loss curve, indicating stable convergence and reduced overfitting.

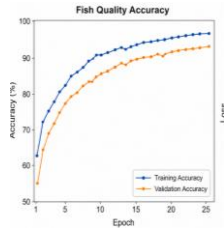


Fig. 9. Accuracy graph of Fish Quality Model

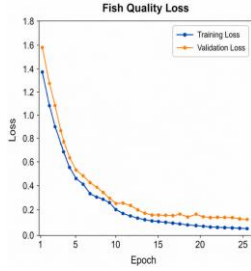


Fig. 10. Loss graph of Fish Quality Model

D. Fish Market Price Forecasting

The price forecasting module was evaluated using regression metrics across multiple models. Random Forest and XGBoost were compared, and an ensemble model was developed.

TABLE IV. – REGRESSION PERFORMANCE (Fish Market)

Model	Model	RMSE	R ²
Random Forest	120.5	180.2	0.89
XGBoost	115.3	172.6	0.91
Ensemble (Proposed)	102.7	160.4	0.93

The ensemble model achieved the best performance with $R^2 = 0.93$, indicating strong predictive capability. Seasonal features and fuel price indicators contributed significantly to improving accuracy.

Fig. 11 shows the trend between predicted and actual prices, while Fig. 12 illustrates model loss convergence.

The price forecasting module was evaluated using regression metrics across multiple models. Random Forest and XGBoost were compared, and an ensemble model was developed.

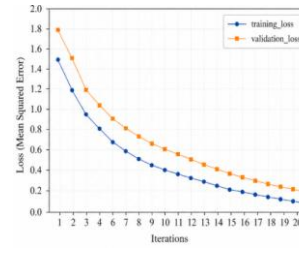


Fig. 11. Accuracy graph of Fish Market Model



Fig. 12. Loss graph of DATCIE Model

The ensemble model achieved the best performance with $R^2 = 0.93$, indicating strong predictive capability. Seasonal features and fuel price indicators contributed significantly to improving accuracy. Fig. 10 shows the trend between predicted and actual prices, while Fig. 11 illustrates model loss convergence.

V. CONCLUSION

This study successfully developed an AI-based multi-functional decision support system for predicting fish zones, fish prices, fish quality, and trip cost, achieving high accuracies of 85%, 93%, 82%, and 83%, respectively. These results highlight the system's effectiveness and potential as a reliable decision support tool for improving fishing operations in Sri Lanka. By enabling accurate zone identification, price forecasting, quality assessment, and cost estimation, this system can significantly contribute to reducing operational inefficiencies and improving fishermen's profitability.

By facilitating data-driven decision-making and optimized resource utilization, the system supports more efficient and sustainable fishing practices. The integration of multiple predictive modules into a unified platform addresses limitations of existing single-function systems and enhances real-world applicability. Future work could focus on incorporating real-time data integration, expanding datasets, improving model accuracy, and deploying the system as a fully functional mobile application for practical use in coastal fishing communities.

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